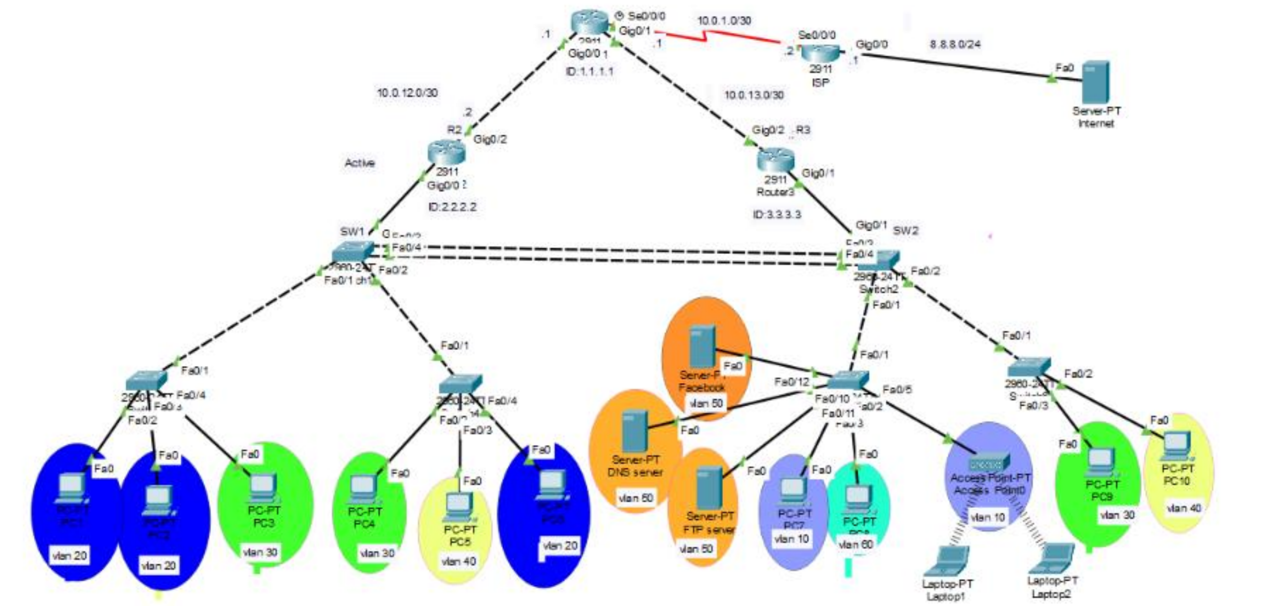
CCNA PROJECT

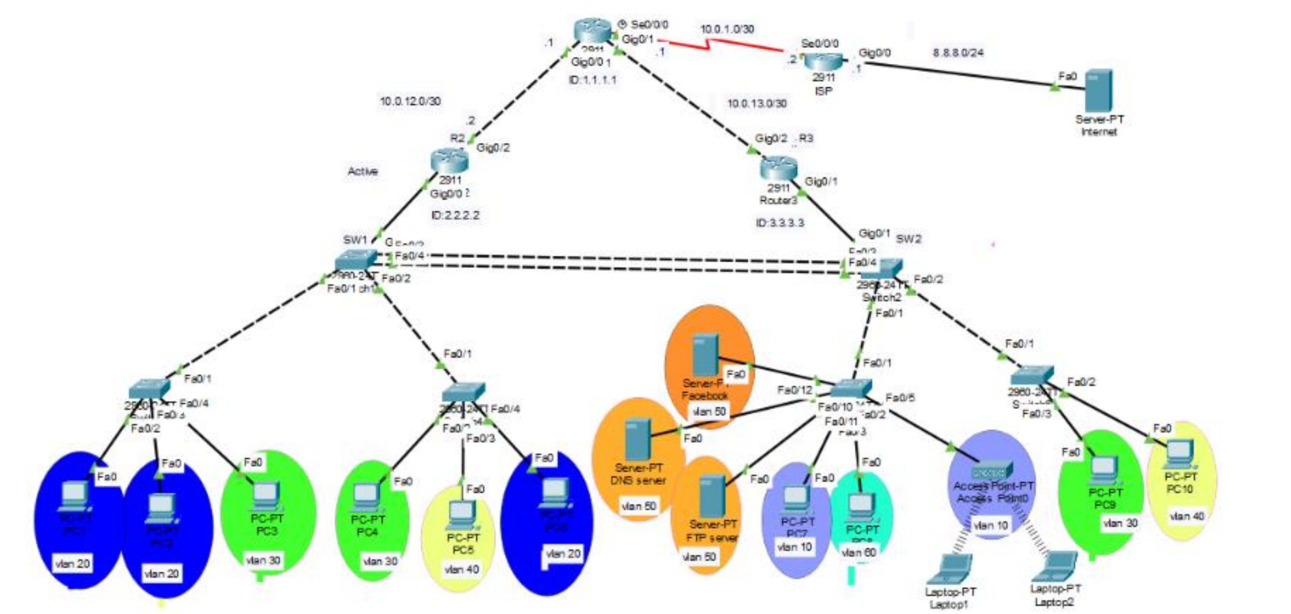
Redundant Enterprise Network Infrastructure with Secure Internet Connectivity



1. Introduction

This project focuses on designing and implementing an enterprise network using Cisco Packet Tracer. The network was developed to provide reliable communication between different departments and services while applying important CCNA networking concepts. The design includes VLAN segmentation, inter-VLAN routing, network redundancy, dynamic routing, DHCP, network security, wireless connectivity, and access to external networks. The main objective of the project is to demonstrate how these technologies can be integrated into a single network to achieve reliable, organized, and secure communication.

2. Topology Overview



Core Devices

R1	Edge router — connects to ISP (2911), NAT overload to internet, OSPF to R2/R3
R2 / R3	Distribution routers — Router-on-a-Stick (VLAN sub-interfaces) + HSRP pair
SW1 / SW2	Core switches — Rapid PVST+, EtherChannel (LACP) between them, trunk to access switches
SW3–SW6	Access switches — access ports to PCs/servers/AP, port security enabled
ISP Router	Simulates internet connectivity, linked to Server-PT Internet

Key Addressing

Link / Segment	Network	Notes
R1 – R2 (Gig0/0)	10.0.12.0/30	OSPF area 0
R1 – R3 (Gig0/1)	10.0.13.0/30	OSPF area 0

Link / Segment	Network	Notes
R1 – ISP (Se0/0/0)	10.0.1.0/30	NAT outside
VLAN 10 – management	192.168.10.0/24	HSRP VIP .1
VLAN 20 – sales	192.168.20.0/24	HSRP VIP .1
VLAN 30 – engineering	192.168.30.0/24	HSRP VIP .1
VLAN 40 – HR	192.168.40.0/24	HSRP VIP .1
VLAN 50 – servers	192.168.50.0/24	DNS/FTP/Web servers, HSRP VIP .1
VLAN 60 – guest	192.168.60.0/24	ACL-restricted, HSRP VIP .1

3. Feature Implementation Summary

Quick-reference status of every feature required for this project, based on the configuration and show-command outputs collected from the devices.

Feature	Status
VLANs	✓ Verified
Access ports	✓ Verified
Trunks	✓ Verified
Native VLAN 99	✓ Verified
EtherChannel / LACP	✓ Verified
Rapid PVST+	✓ Verified
Router-on-a-Stick	✓ Verified
DHCP	✓ Verified
HSRP	✓ Verified
OSPF	✓ Verified
Port Security	✓ Verified
ACL	✓ Verified
AP + Guest VLAN	✓ Verified
Port Security + BPDU Guard	✓ Verified

4. Feature Details & Verification

4.1. VLANs

Purpose: Logical segmentation of the switched network into separate broadcast domains, one per department/function.

Applied to: SW1 & SW2 (and access switches SW3–SW6): VLAN 10 (management), 20 (sales), 30 (engineering), 40 (HR), 50 (servers), 60 (guest), 99 (native), 999 (parking-lot).

Verification command: `show vlan brief`

```
VLAN Name                    Status
1    default                 active
10   management               active
20   sales                    active
30   engineering              active
40   HR                       active
50   servers                  active
60   guest                    active
99   native                   active
999  parking-lot              active
```

4.2. Access Ports

Purpose: Connect end devices (PCs, servers, AP) to a single, statically assigned VLAN.

Applied to: e.g. SW3 Fa0/2–Fa0/3 (VLAN 20 - sales), Fa0/4 (VLAN 30 - engineering); similar static access ports across SW3–SW6 for VLANs 10/20/30/40/50/60.

Verification command: `show interfaces switchport`

```
Name: Fa0/2
Administrative Mode: static access
Operational Mode: static access
Access Mode VLAN: 20 (sales)
```

4.3. Trunks

Purpose: Carry multiple VLANs over a single link between switches using 802.1Q tagging.

Applied to: SW1 & SW2: Po1, Fa0/1, Fa0/2, Gig0/1 — all set to trunk mode carrying VLANs 1,10,20,30,40,50,60,99,999.

Verification command: `show interfaces trunk`

```
Port   Mode  Encapsulation  Status     Native vlan
Po1    on    802.1q         trunking   99
Fa0/1  on    802.1q         trunking   99
Fa0/2  on    802.1q         trunking   99
Gig0/1 on    802.1q         trunking   99
```

4.4. Native VLAN 99

Purpose: Untagged VLAN on trunk links, deliberately separated from data VLANs to prevent VLAN-hopping attacks.

Applied to: Configured on every trunk interface on SW1, SW2, and the trunk ports on SW3–SW6 (switchport trunk native vlan 99).

Verification command: `show interfaces trunk`

```
Native vlan column = 99 on all trunk ports (Po1, Fa0/1, Fa0/2, Gig0/1).
```

4.5. EtherChannel / LACP

Purpose: Bundles multiple physical links into one logical link (Port-channel) for redundancy and load balancing, using the dynamic LACP protocol.

Applied to: SW1 Fa0/3 + Fa0/4 and SW2 Fa0/3 + Fa0/4, bundled into Port-channel 1 (channel-group 1 mode active).

Verification command: `show etherchannel summary`

```
Group  Port-channel  Protocol  Ports
1      Po1(SU)       LACP      Fa0/3(P) Fa0/4(P)
```

4.6. Rapid PVST+

Purpose: Fast-converging spanning-tree instance per VLAN, prevents Layer 2 loops on the redundant SW1–SW2–access-switch topology.

Applied to: Enabled globally on SW1/SW2/SW3-6. SW1 is root for VLANs 10/20/30 (priority 24576); SW2 is root for VLANs 40/50/60 (priority 24576) — load balances root role across the two core switches.

Verification command: `show spanning-tree summary`

```
Switch is in rapid-pvst mode
Root bridge for: management sales engineering   (SW1)
Root bridge for: HR servers guest                (SW2)
```

4.7. Router-on-a-Stick

Purpose: Inter-VLAN routing performed on a single physical trunk link using 802.1Q sub-interfaces, one per VLAN.

Applied to: R2 (Gig0/0.10–.60, .99 native) and R3 (Gig0/1.10–.60, .99 native), trunked to SW1/SW2.

Verification command: `show ip interface brief`

```
Interface              IP-Address     Status  Protocol
GigabitEthernet0/0.10  192.168.10.2   up      up
GigabitEthernet0/0.20  192.168.20.2   up      up
GigabitEthernet0/0.30  192.168.30.2   up      up
GigabitEthernet0/0.40  192.168.40.2   up      up
GigabitEthernet0/0.50  192.168.50.2   up      up
GigabitEthernet0/0.60  192.168.60.2   up      up
```

4.8. DHCP

Purpose: Automatic IP address, default gateway, and DNS assignment for end devices, one pool per VLAN.

Applied to: Configured on R2: pools vlan10–vlan60 (network + default-router + dns-server 192.168.50.10), with excluded static ranges .1–.20 per subnet.

Verification command: `show ip dhcp binding`

```
IP address       Hardware address    Type
192.168.10.21    00E0.B086.9B0D       Automatic
192.168.20.21    0040.0BD9.9163       Automatic
192.168.30.21    000B.BE06.9B19       Automatic
192.168.40.21    0001.4359.144A       Automatic
192.168.60.21    0060.3E2E.A009       Automatic
```

4.9. HSRP

Purpose: First-Hop Redundancy Protocol — provides a shared virtual gateway IP so hosts keep connectivity if the active router fails.

Applied to: R2 = Active (priority 150, preempt) and R3 = Standby (priority 100, preempt) for VLANs 10/20/30/40/50/60. Virtual IPs end in .1 on every subnet.

Verification command: `show standby brief`

```
Interface  Grp  Pri  P  State    Active         Standby        Virtual IP
Gig0/0.10  10   150  P  Active   local          192.168.10.3   192.168.10.1
Gig0/0.20  20   150  P  Active   local          192.168.20.3   192.168.20.1
...(same pattern for VLANs 30/40/50/60)
```

4.10. OSPF

Purpose: Dynamic link-state routing protocol distributing routes between R1, R2, and R3, and for the VLAN subnets on R2/R3.

Applied to: Single process (router ospf 10), area 0. R1 router-id auto, R2 router-id 2.2.2.2, R3 router-id 3.3.3.3. R2 and R3 are tuned with ip ospf priority 150 on VLAN sub-interfaces so they win DR election over each other consistently.

Verification command: `show ip ospf neighbor`

```
Neighbor ID   Pri  State      Address        Interface
2.2.2.2       1    FULL/DR    10.0.12.2      GigabitEthernet0/0   (on R1)
3.3.3.3       1    FULL/DR    10.0.13.2      GigabitEthernet0/1   (on R1)
```

4.11. Port Security

Purpose: Restricts each access port to a limited number of learned MAC addresses (sticky), protecting against rogue devices / MAC flooding.

Applied to: Access ports on SW3–SW6 (e.g. Fa0/2): maximum 1 MAC address, sticky learning, violation mode Restrict.

Verification command: `show port-security interface f0/2`

```
Port Security        : Enabled
Port Status          : Secure-up
Violation Mode       : Restrict
Maximum MAC Addresses : 1
Sticky MAC Addresses  : 1
Security Violation Count : 0
```

4.12. ACL

Purpose: Extended IP ACL used as a security policy to isolate the guest VLAN from the servers VLAN.

Applied to: Applied inbound on R2 Gig0/0.60 (guest VLAN sub-interface): ip access-group guest-filter in.

Verification command: `show access-lists`

```
Extended IP access list guest-filter
 10 deny ip 192.168.60.0 0.0.0.255 192.168.50.0 0.0.0.255 (24 matches)
 20 permit ip any any (6056 matches)
```

4.13. AP + Guest VLAN

Purpose: Wireless access point providing connectivity for guest laptops, placed in an isolated VLAN with restricted access to internal servers.

Applied to: Applied to: AccessPoint0 with Laptop1 and Laptop2, connected on VLAN 10 (management) **Verification command:** `ping from Laptop1/Laptop2 to a server IP (expected: success, since VLAN 10 is not restricted by the ACL)`

```
C:\>ping 192.168.50.11

Pinging 192.168.50.11 with 32 bytes of data:

Reply from 192.168.50.11: bytes=32 time=25ms TTL=127
Reply from 192.168.50.11: bytes=32 time=7ms TTL=127
Reply from 192.168.50.11: bytes=32 time=20ms TTL=127
Reply from 192.168.50.11: bytes=32 time=48ms TTL=127

Ping statistics for 192.168.50.11:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
```

```
Minimum = 7ms, Maximum = 48ms, Average = 25ms
```

4.14. PortFast

Purpose: Bypasses the STP listening/learning states on access (edge) ports so end devices get connectivity immediately after link-up.

Applied to: Intended for the same access ports as Port Security (PCs, servers, AP) on SW3–SW6.

Verification command: `show spanning-tree interface f0/2 detail`

```
Port 2 (FastEthernet0/2) of VLAN0030 is designated forwarding
Port path cost 19, Port priority 128, Port Identifier 128.2
Designated root has priority 24606, address 0005.5E48.688B
Designated bridge has priority 32798, address 0090.2B4B.3157
Designated port id is 128.2, designated path cost 19
Timers: message age 16, forward delay 0, hold 0
Number of transitions to forwarding state: 1
The port is in the portfast mode
Link type is point-to-point by default
```

4.15. BPDU Guard

Purpose: Automatically error-disables a PortFast port if it receives a BPDU, protecting against accidental loops from a rogue switch.

Applied to: Intended for the same PortFast-enabled access ports on SW3–SW6.

Verification command: `show running configuration`

```
interface FastEthernet0/2
switchport access vlan 20
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security violation restrict
switchport port-security mac-address sticky 0002.1611.1B24
spanning-tree portfast
spanning-tree bpduguard enable
! 9 vlans 24 0 0 12 36
```

5. Network Connectivity Testing

Connectivity tests were performed to verify that the implemented network functions correctly and that communication is available between the different network segments. Ping tests were used to verify connectivity between end devices and their default gateways, between different VLANs, between internal devices and servers, and between the internal network and the simulated Internet. The tests also included the ACL configuration to confirm that restricted traffic from the guest VLAN to the server VLAN is blocked as intended. The successful and failed ping results below provide evidence that routing, VLAN communication, gateway redundancy, server connectivity, Internet connectivity, and traffic filtering are operating according to the project design.

Ping Test 1 – Default Gateway:

```
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping Test 2 – Inter-VLAN Connectivity:

```
C:\>ping 192.168.30.22

Pinging 192.168.30.22 with 32 bytes of data:

Reply from 192.168.30.22: bytes=32 time<1ms TTL=127
Reply from 192.168.30.22: bytes=32 time<1ms TTL=127
Reply from 192.168.30.22: bytes=32 time<1ms TTL=127
Reply from 192.168.30.22: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping Test 3 – Server Connectivity:

```
C:\>ping 192.168.50.11

Pinging 192.168.50.11 with 32 bytes of data:

Reply from 192.168.50.11: bytes=32 time<1ms TTL=127
Reply from 192.168.50.11: bytes=32 time=13ms TTL=127
Reply from 192.168.50.11: bytes=32 time<1ms TTL=127
Reply from 192.168.50.11: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.50.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 13ms, Average = 3ms
```

Ping Test 4 – Internet Connectivity:

```
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time=1ms TTL=125
Reply from 8.8.8.8: bytes=32 time=1ms TTL=125
Reply from 8.8.8.8: bytes=32 time=1ms TTL=125
Reply from 8.8.8.8: bytes=32 time=1ms TTL=125

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

Ping Test 5 – ACL Verification (Valan 60 Can’t reach Servers):

```
C:\>ping 192.168.50.10

Pinging 192.168.50.10 with 32 bytes of data:

Reply from 192.168.60.2: Destination host unreachable.
Reply from 192.168.60.2: Destination host unreachable.
Reply from 192.168.60.2: Destination host unreachable.
Reply from 192.168.60.2: Destination host unreachable.

Ping statistics for 192.168.50.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

6. Conclusion

The completed project successfully demonstrates the implementation of an enterprise network using Cisco technologies. The network combines VLANs, trunking, EtherChannel, Rapid PVST+, HSRP, OSPF, DHCP, ACLs, and network security features to provide segmentation, redundancy, routing, and controlled access. Connectivity and configuration tests were performed to verify the operation of the main network functions. Overall, the project provided practical experience in designing, configuring, and troubleshooting a reliable enterprise network environment.